LINEAR ALGEBRA · LESSON 12

Graphs, Networks, Incidence Matrices

Turn a network of prices into a matrix. Loops become the **left null space**, and a loop that doesn't add to zero is an **arbitrage**.

nodes − edges + loops = 1 | $A^T y = 0$ | MIT 18.06, Lecture 12

Contracts and spreads

Nodes: Nifty spot, Oct, Nov, Dec futures (20,000 / 20,100 / 20,250 / 20,400).

Edges: quoted spreads, e.g. Oct − spot = 100.

Incidence matrix: one row per edge, -1 at the start node, $+1$ at the end node. Then

$$Ax = \text{the spread on every edge.}$$

THE NULL SPACE

Only relative prices matter

$Ax = 0 \Rightarrow$ all prices equal: $N(A) = c(1, 1, 1, 1)$.

Add 50 points to every price and **no spread changes.**

> GROUND A NODE
>
> Fix spot = 20,000 (the numeraire) and the spreads decide every other price. Rank = $n - 1$.

LOOPS

Positions that cancel

Hold y_i lots of spread i. Then $A^T y$ = net position in each contract.

$A^T y = 0$ (Kirchhoff's law): flat in every contract. Its solutions are **loops**:

$$y = (1, 1, -1, 0, 0)$$

buy spot→Oct, buy Oct→Nov, sell spot→Nov: net zero in every contract.

COUNTING

Euler's formula

Loops $= \dim N(A^T) = m - (n-1)$. So

$$\text{nodes} - \text{edges} + \text{loops} = 1$$

4 contracts, 5 spreads: $4 - 5 + 2 = 1$ ✓. Two independent loops.

THE TEST

No arbitrage = every loop sums to 0

Quotes come from real prices $\Leftrightarrow e \in C(A) \Leftrightarrow y^T e = 0$ for every loop (Lesson 10's test).

Consistent: $100 + 150 - 250 = 0$ ✓

Mispriced spot→Nov at 240: $100 + 150 - 240 = 10$ ✗

THE TRADE

Trade the loop

Buy the direct spot→Nov spread at 240, sell spot→Oct and Oct→Nov (worth 250).

> RESULT
>
> Net 0 lots in every contract ($A^T y = 0$), **10 points** locked in.

Currency triangles work the same way with logs: $83 \times 1.08 = 89.64$, so a ₹90/€ quote is 0.4% off.

RECAP

Four subspaces of a network

Space	Dim	Meaning
$N(A)$	1	shift all prices
$C(A)$	$n-1$	consistent quotes
$C(A^T)$	$n-1$	positions from spreads
$N(A^T)$	$m-n+1$	loops: the arbitrage test

QUANT FROM FIRST PRINCIPLES

The full interactive lesson is free

A spread checker, a network lab that finds loops and arbitrage, and worked currency and lot-size examples:

learn.krisarj.com

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